

Learning Latent Program Spaces through Neural DSL Navigation*

Julien Livet

Independent Researcher

JULIEN.LIVET@FREE.FR

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Abstract

This paper presents an exploratory neuro-symbolic approach for learning a latent representation of programs expressed in a Domain-Specific Language (DSL) inspired by ARC-style grid transformations. The system combines Graph Neural Networks (GNNs), Convolutional Neural Networks (CNNs), and Transformers within a unified architecture trained with an iterative cost-guided objective. Preliminary empirical observations indicate that structured neighbourhoods emerge *spontaneously* in the embedding space, with programs clustering by semantic function (rotations, mirrors, reductions, concatenations). Furthermore, the model exhibits non-trivial conceptual transitions during program search, including the formation of anchors and star-shaped exploration patterns. These results suggest that the proposed method offers a promising direction for explainable, data-efficient neuro-symbolic program synthesis.

1. Introduction

The field of Neuro-Symbolic Artificial Intelligence has experienced rapid growth, with research efforts concentrated in learning and inference (63%), logic and reasoning (35%), and knowledge representation (44%) (Colelough et al., 2025). However, significant gaps remain in **explainability** (28%) and **meta-cognition** (5%) – areas that are critical for deploying AI systems in high-stakes domains such as industrial automation, healthcare, and formal verification.

Program synthesis – the automatic construction of programs from specifications – remains a challenging task due to the combinatorial explosion of the search space. Pure neural approaches lack interpretability, while pure symbolic methods struggle with scalability and noise tolerance.

This paper investigates a hybrid approach where: (i) a **DSL** provides a structured symbolic vocabulary; (ii) a **neural architecture** (GNN + CNN + Transformer) learns a continuous latent space of programs; and (iii) an **iterative cost-guided search** navigates this space using a multi-component heuristic.

We demonstrate that this architecture leads to the **spontaneous emergence of a semantic map** of the program space, where programs with similar effects cluster together – without any explicit semantic supervision.

* Code available at <https://github.com/Julien-Livet/aicpp>

2. Related Work

Neuro-Symbolic AI. Recent systematic reviews (Colelough et al., 2025) highlight a growing interest in integrating symbolic reasoning with deep learning. Approaches range from logically constrained neural networks (Mao et al., 2019) to graph-based knowledge representation (Schlichtkrull et al., 2018). However, most works focus on classification or reasoning tasks, not on program synthesis.

Program Synthesis. Classical approaches use deductive synthesis (Gulwani, 2011), while modern methods leverage large language models (LLMs) for code generation (Chen et al., 2021). LLMs lack explicit program structure and are difficult to constrain. DreamCoder (Ellis et al., 2021) introduced wake-sleep library learning for program induction from examples, and STITCH (Bowers et al., 2023) further refined top-down abstraction in DSL spaces. Our approach complements these by learning a *continuous* latent map of the program space rather than a discrete library.

Latent Program Spaces. Several works have explored continuous representations of programs through variational autoencoders (Bunel et al., 2017) or by training transformers on program traces (Zaremba and Sutskever, 2015). However, these often produce unstructured latent spaces. Our work shows that structured semantic maps can emerge when using a typed DSL combined with a multi-component cost function.

ARC and ARC-AGI. The Abstraction and Reasoning Corpus (ARC) (Chollet, 2019) is a benchmark for measuring general intelligence. Neuro-symbolic approaches to ARC remain rare, with most top performers relying on combinatorial search (Hodel, 2024). This work positions itself within this niche.

3. The C++ Domain-Specific Language

The proposed approach relies on a domain-specific language (DSL) implemented in C++. The DSL defines the executable vocabulary available to the program synthesis system. Rather than generating unrestricted source code, the model therefore operates over a constrained space of semantically meaningful programs.

3.1. DSL Design Principles

The DSL is designed around three main principles.

First, each primitive represents a meaningful transformation in the target domain. The vocabulary is therefore considerably smaller than a general purpose programming language.

Second, primitives are strongly typed according to their input and output domains. This allows syntactically invalid compositions to be rejected during program construction.

Third, DSL programs are directly executable by the C++ engine. A generated program can consequently be evaluated against the training examples without requiring an additional interpretation or compilation step.

These properties constrain the search space while preserving an explicit representation of the transformations performed by a synthesized program.

3.2. Typed Primitives

Each DSL primitive is associated with a fixed signature describing its input and output types. For example, an image transformation may be represented as

```
Image mirror(Image input)
```

while a parameterized transformation can be represented as

```
Image upscale(Image input, Integer factor)
```

The type system consequently constrains the combinations of primitives that can be considered during synthesis. This is particularly important when the program space contains primitives with heterogeneous signatures.

In the implementation, primitive signatures are exposed to the synthesis engine through the C++ representation of the DSL. The model does not have to learn these typing constraints from the data alone: they are explicitly encoded in the program-generation mechanism.

3.3. Program Representation

A DSL program is represented as a composition of primitive applications. For example, a transformation can be expressed as

```
hconcat(vmirror(I), cmirror(I))
```

where I denotes the input grid.

Nested expressions naturally form a tree structure. This representation is used both for program generation and for constructing the graph representation provided to the neural model.

The tree representation also makes the synthesized programs directly inspectable by a human observer. Unlike a latent representation in which the reasoning process is not directly observable, a generated DSL program explicitly exposes the sequence of transformations selected by the system.

3.4. Executable Semantics

The C++ engine provides the operational semantics of the DSL. Given a valid program and an input grid, the engine recursively evaluates the corresponding expression tree and produces an output grid.

The same execution mechanism is used during program search to evaluate candidate programs against the training examples. A candidate program can therefore be assigned a cost according to its discrepancy with the expected outputs.

Let P denote a synthesized program and let $(X_i, Y_i)_{i=1}^N$ denote the training pairs. The program cost can abstractly be written as

$$C(P) = \sum_{i=1}^N D(P(X_i), Y_i),$$

where D is the discrepancy measure implemented by the evaluation engine.

A value of $C(P) = 0$ indicates that the synthesized program reproduces all training outputs exactly.

3.5. DSL-Constrained Program Search

The DSL induces a structured program space

$$\mathcal{P}_{\text{DSL}} = \{P \mid P \text{ is a well-typed composition of DSL primitives}\}.$$

Program synthesis is consequently performed over $\mathcal{P}_{\text{DSL}}$ rather than over arbitrary programs.

For a maximum synthesis depth d , the search space can be viewed as the subset

$$\mathcal{P}_{\text{DSL}}^{(\leq d)} = \{P \in \mathcal{P}_{\text{DSL}} \mid \text{depth}(P) \leq d\}.$$

This depth constraint provides a direct mechanism for controlling the computational complexity of synthesis.

The search engine evaluates candidate programs and retains programs that improve the current cost. Consequently, the search trajectory provides an explicit record of the successive hypotheses considered by the system.

3.6. Integration with the Neural Model

The DSL is not used solely as an execution language. It also constitutes the interface between neural prediction and symbolic program synthesis.

The neural model predicts DSL tokens and program structures, while the symbolic engine validates and executes the resulting programs. Candidate programs can subsequently be represented as program graphs and combined with their observed execution costs before being provided to the model.

This creates a feedback loop between the neural and symbolic components:

1. the neural model proposes a DSL program;
2. the symbolic engine validates and executes the program;
3. the resulting program is evaluated against the training examples;
4. the execution cost provides feedback about the quality of the candidate;
5. the resulting information is incorporated into the subsequent program-generation process.

The architecture therefore combines learned program generation with explicit symbolic execution.

3.7. Interpretability of Synthesized Programs

An important consequence of the DSL representation is that successful or partially successful hypotheses remain expressed as executable symbolic programs.

For example, the search process can produce expressions such as

```
trim(trim(I))
```

or

```
upscale(dedupe(I), TEN).
```

Such expressions directly expose the transformations selected by the system. They can therefore be inspected, executed independently, compared with alternative hypotheses, and used to analyse the behaviour of the synthesis process.

This property is particularly relevant to ARC-like tasks, where understanding the transformation applied to an input grid is an important part of analysing a solution.

3.8. Domain Adaptability

The DSL is intended to separate the general synthesis mechanism from the domain-specific vocabulary.

Changing the target domain primarily requires defining a new set of typed primitives and their executable semantics. The neural and symbolic synthesis components can then operate over this new vocabulary.

This design suggests that the same general architecture could potentially be applied to domains other than grid transformations, provided that an appropriate DSL can be defined.

4. Method

4.1. Domain-Specific Language

We use a typed DSL derived from the Hodel DSL for ARC (Hodel, 2024), containing primitives such as rotations (`rot90`, `rot180`, `rot270`), mirrors (`hmirror`, `vmirror`, `cmirror`, `dmirror`), reductions (`tophalf`, `bottomhalf`, `lefthalf`, `righthalf`), concatenations (`hconcat`, `vconcat`), rescaling (`upscale`, `downscale`, `vupscale`), and value transformations (`replace`, `switch`, `compress`, `trim`).

All programs are typed and must satisfy syntactic constraints enforced by a `TreeState` mask: at each decoding step, only tokens whose type is compatible with the current position in the Abstract Syntax Tree (AST) are allowed. This guarantees that every generated program is syntactically valid by construction.

4.2. Architecture

The model consists of three encoders and one constrained decoder, illustrated schematically in Fig. 1.

ARC Grid Encoder (CNN). Each pair $(I_{\text{in}}, I_{\text{out}})$ is encoded by a four-layer CNN ($64 \rightarrow 64 \rightarrow 128 \rightarrow 128$ channels, kernel size 3×3 , padding 1) with attention pooling over N

pairs. Each output grids of M candidate programs are also encoded. The input tensor concatenates one-hot colour channels, the pixel-wise difference map, and a validity mask. Output: $z_{\text{grid}} \in \mathbb{R}^{256}$.

Program Encoder (GNN). A DSL program is parsed into an AST and passed through three GCNConv layers (Schlichtkrull et al., 2018) with mean pooling. Output: $z_{\text{prog}} \in \mathbb{R}^{256}$.

Cost Encoder (MLP + Attention). The cost tensor of shape $[N_{\text{grids}}, 5]$ – comprising grid-size cost, bounding-box cost, pixel-overlap cost, value cost, and total cost – is log-normalised ($\log_{1p}$), passed through a two-layer MLP, and aggregated with attention pooling. Output: $z_{\text{cost}} \in \mathbb{R}^{256}$.

Fusion and Decoder. z_{prog} and z_{cost} are summed element-wise. Cross-attention between the grid context and program embeddings produces a unified context vector z_{ctx} . A Transformer decoder (4 layers, 8 heads, $d_{\text{model}} = 256$) generates new DSL programs autoregressively, with the **TreeState** mask applied at each step.

Training objective. When a sampled candidate matches or improves upon the current best cost in the program pool, the decoder is trained to reproduce its own token sequence (self-imitation); otherwise, it is trained against the best-known reference program from the search trajectory. An entropy regularisation term prevents premature collapse of the output distribution.

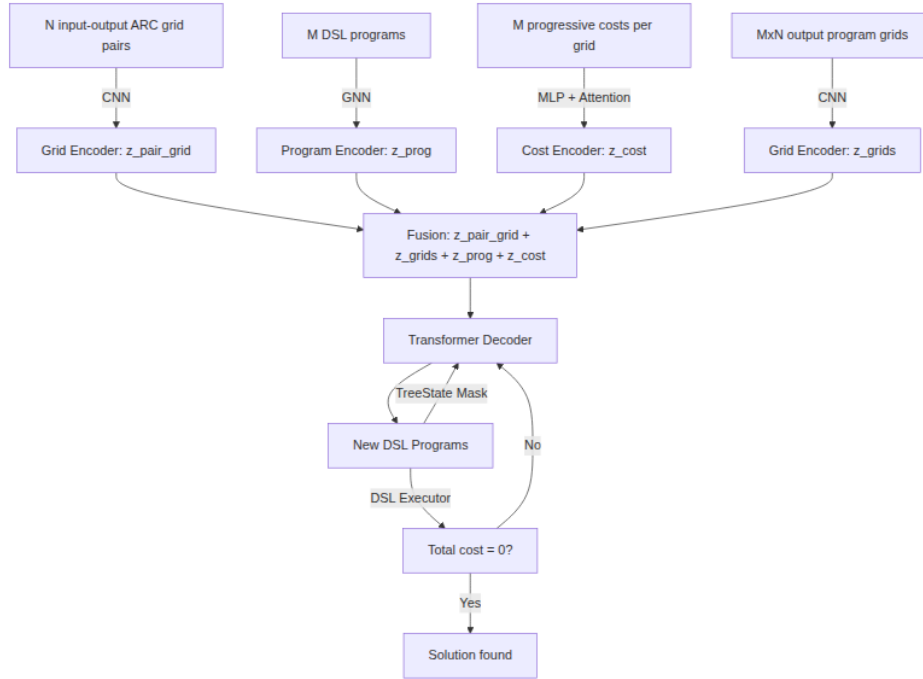


Figure 1: Architecture overview. Three encoders (CNN for grids, GNN for programs, MLP for costs) produce a unified context vector consumed by a **TreeState**-constrained Transformer decoder.

4.3. Data Generation

We generate 10,000 valid DSL programs by random walks in the typed program space, using the `TreeState` mask to guarantee syntactic validity. For each program we compute its output on 3–6 randomly generated ARC grid pairs (up to 30×30), producing the cost vector used for training.

4.4. Inference: Iterative Cost-Guided Search

Starting from the identity program `I`, the model iteratively: (1) encodes the current candidate programs and their costs; (2) proposes new programs via beam search (width 20); (3) executes proposed programs and computes actual costs; (4) retains the best $M = 50$ candidates and repeats.

We observe that this process often follows a **star-shaped exploration pattern**: the model tests multiple variations around a promising anchor program before moving to a different region of the program space. This behavior provides a natural setting for studying how the training regime interacts with search, since multiple partially explored trajectories may coexist and contribute experience to subsequent model updates.

In addition to the search procedure itself, the training regime affects how experience is incorporated into the model. We consider both synchronous and asynchronous training configurations. In the synchronous setting, candidate generation and learning updates are organized around a single search trajectory, so that successive variants of a candidate can be generated under a relatively stable model state. In the asynchronous actor–learner setting, several search trajectories may be explored concurrently by actors while a central learner continuously updates the model. Consequently, experiences generated from different trajectories can be incorporated in an interleaved order: for example, candidates from trajectories A , B , and C may contribute to successive updates before later variants A' , B' , and C' are processed.

This asynchronous organization exposes the model to a more interleaved stream of search experiences and allows model updates to influence trajectories that are already in progress. The resulting interaction between exploration and learning is therefore different from the more sequential exposure obtained in the synchronous setting. We investigate whether this interleaving contributes to more effective navigation of the program space, in particular by enabling experience acquired on one search trajectory to influence the exploration of others.

5. Experimental Setup

Hardware. 28 CPU cores for data generation and cost evaluation; one NVIDIA RTX 2000 GPU for model training and inference.

Dataset. 10,000 unique DSL programs with syntactic constraints, up to 128 tokens, evaluated on grids up to 30×30 .

Training. AdamW optimiser ($\text{lr} = 10^{-4}$), batch size of 1 task with $M = 50$ candidate programs per iteration, online training over several hours.

Visualisation. Embeddings are extracted from the trained GNN encoder for 100 programs sorted by length. t-SNE (perplexity=30) and UMAP (cosine metric) projections are computed with standard parameters.

6. Results

6.1. Emergent Semantic Clustering

Both t-SNE (Fig. 2) and UMAP (Fig. 3) projections reveal a clearly structured latent space. Rotations (`rot90`, `rot180`, `rot270`) form a dense, well-separated cluster; mirrors occupy another distinct region; reductions are grouped together; concatenations are separated by axis; and value transformations lie in a transitional area.

Importantly, **no semantic label was provided during training**. The structure emerged solely from the cost-guided objective and the DSL type constraints.

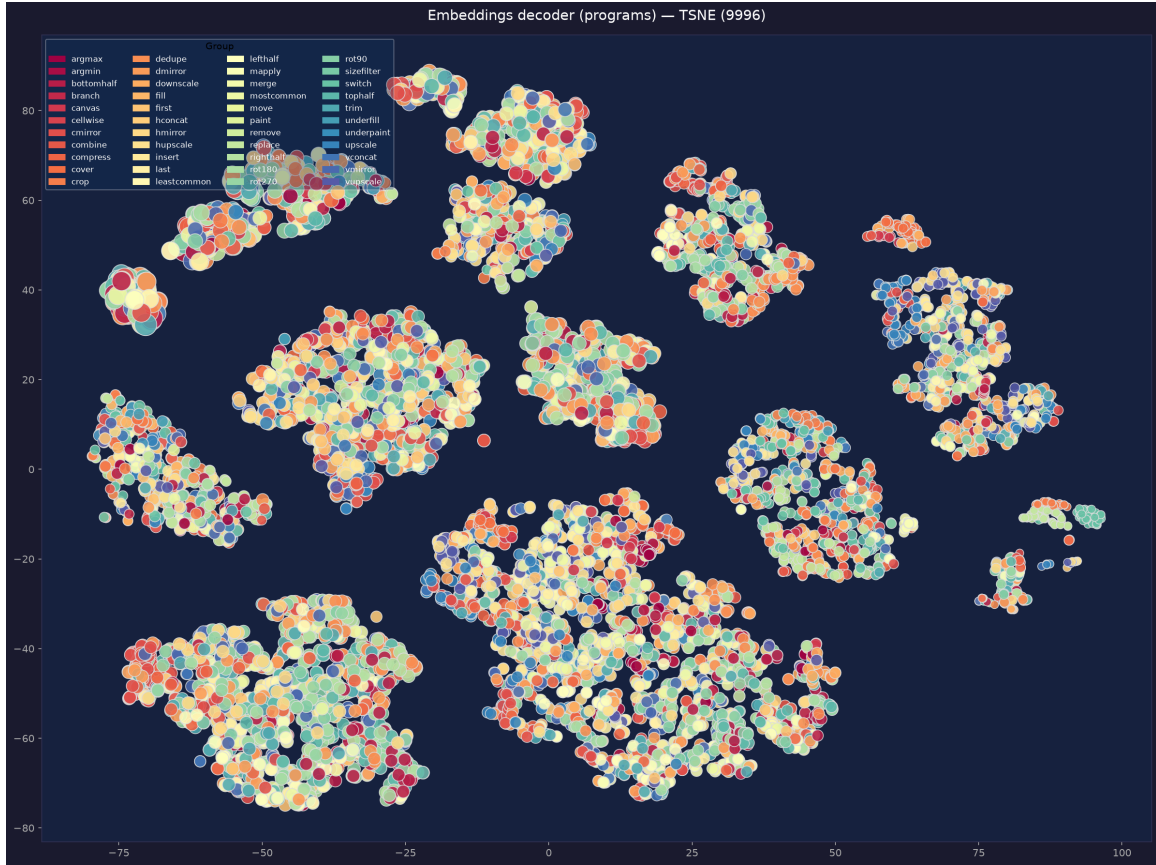


Figure 2: t-SNE visualisation of 10000 DSL program embeddings. Colour encodes the root primitive; point size encodes AST depth. Semantically related primitives cluster spontaneously.

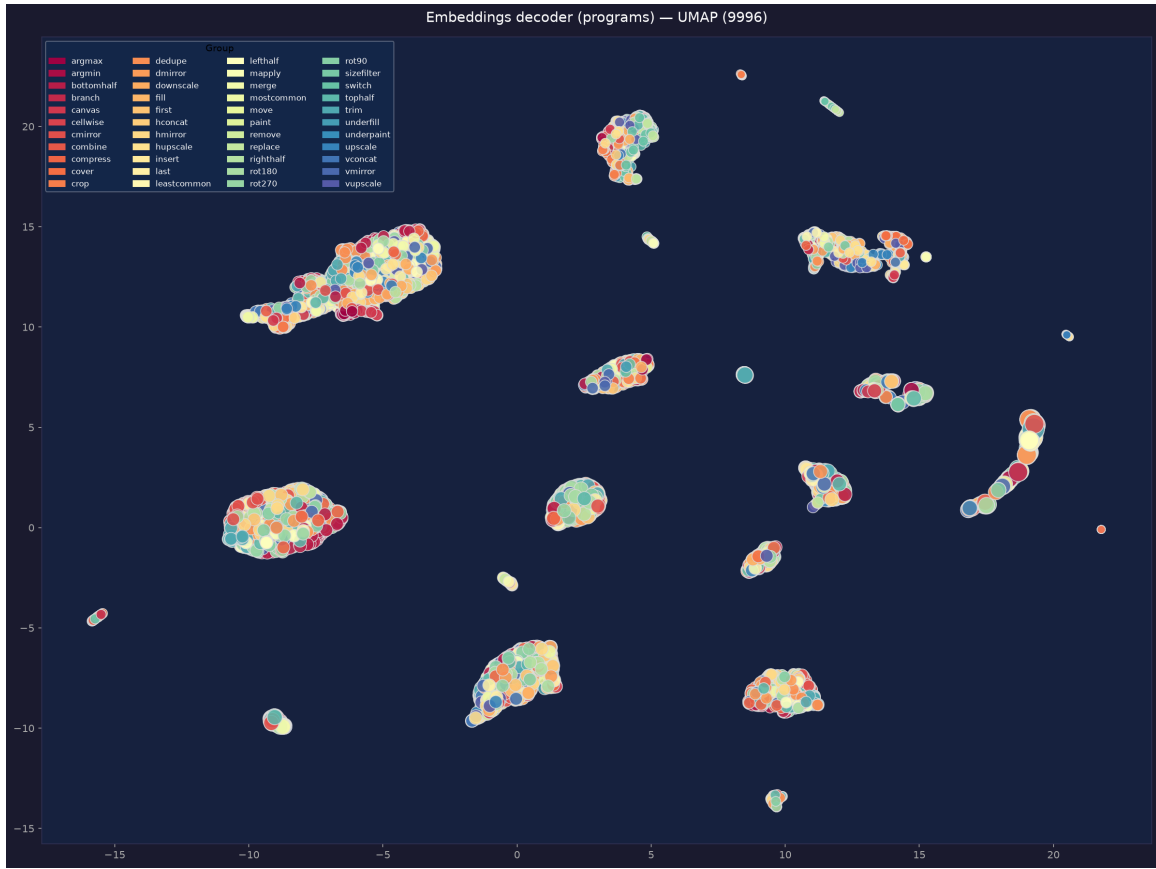


Figure 3: UMAP visualisation of 10000 DSL program embeddings.

6.2. Compositionality and Continuity

Programs composed of two primitives (e.g., `tophalf(trim(I))`) are located midway between the clusters of their components, suggesting that the embedding space is **compositional**: the vector of a composite program approximates the sum of its parts. Furthermore, a continuous path exists from `rot90(I)` to `rot180(I)` to `rot270(I)`, indicating that the model learned a smooth rotational manifold.

6.3. Search Dynamics

During iterative search, the model exhibits **anchor-based exploration**: it first discovers a promising primitive (e.g., `tophalf(I)`), then explores its neighbourhood systematically (e.g., `tophalf(rot90(I))`, `tophalf(vmirror(I))`), and finally jumps to a better anchor. An example trace is given in Appendix B.1. This behaviour resembles **meta-cognition**: the model adapts its search strategy based on observed progress.

6.4. Comparison with LLM-based Baselines

Table 1 compares our approach with four LLM baselines on the ARC-AGI-2 training and evaluation sets using the same iterative pipeline.

Table 1: Comparison of iterative DSL-synthesis approaches on ARC-AGI-2 tasks (success on training subset 120 tasks and evaluation tasks evaluated).

Model	Architecture	Resources	Training (%)	Evaluation (%)
GPT-5	LLM (general)	Cloud API	56.7	2.5
gpt-oss:120b	LLM (open-weight)	High-end GPU	31.7	0.0
Ours	GNN+CNN+Transformer	Local RTX 2000	10.8	0.0
deepseek-r1:8b	LLM (open-weight)	Local RTX 2000	0.0	0.0
llama3:8b	LLM (open-weight)	Local RTX 2000	0.0	0.0
mistral:7b	LLM (open-weight)	Local RTX 2000	0.0	0.0

7. Discussion

7.1. Why Does the Latent Space Self-Organise?

We hypothesise that three factors are crucial. First, the typed DSL with `TreeState` masking eliminates syntactically invalid programs, removing noise from the learning signal. Second, the multi-component cost function (size, bounding box, pixel overlap, value) provides a rich, smooth gradient signal. Third, training on *improvement trajectories* – rather than imitating a fixed dataset – encourages the model to learn a generalised notion of program quality.

7.2. Advantages over Existing Approaches

Compared to LLM-based program synthesis, our approach (i) is explicitly constrained to produce only syntactically valid outputs; (ii) provides interpretable, visualisable embeddings; and (iii) requires no massive pre-training corpus. Compared to pure symbolic search, it learns a distance metric over programs that guides exploration and generalises across tasks.

7.3. Limitations

Scalability. Currently validated on grid transformations; extension to other DSLs (e.g., Lean tactics) remains future work. **Quantitative evaluation.** We evaluated the model on the public ARC-AGI-2 evaluation set as well as on the Kaggle ARC Prize competition leaderboard. In both cases, the success rate was 0%, contrasting sharply with the 10.8% recorded on the training set (the model being insufficiently trained). **Generalization.** Whether the semantic map transfers to unseen task distributions without fine-tuning requires further study.

8. Perspectives

The results presented in this work suggest several directions for extending the proposed approach beyond the ARC domain. A central property of the architecture is the separation between the neural program-generation mechanism and the domain-specific executable language. This separation makes it possible to consider the definition of alternative DSLs for domains with different structures, constraints, and objectives.

8.1. Chess

Chess provides a particularly interesting test case for this approach. A chess-specific DSL could represent the state of a game as a typed board and provide primitives corresponding to legal chess operations, such as selecting a piece, computing its legal moves, moving a piece, capturing an opponent piece, or performing special moves such as castling and en passant.

The resulting programs would represent explicit sequences of chess actions. The symbolic execution engine could enforce the rules of the game and reject illegal states or moves. The neural component could then learn to generate structured sequences of legal operations rather than directly predicting unconstrained board representations.

Such an extension would also make it possible to investigate whether the proposed neural-symbolic interaction can learn increasingly long sequences of dependent decisions. In contrast with ARC tasks, where a program generally maps an input grid to an output grid, chess would introduce a dynamic state space in which each operation modifies the state on which subsequent operations depend.

8.2. Computer-Aided Design

Computer-aided design (CAD) represents another potential application domain. A CAD-oriented DSL could expose geometric primitives and operations such as creating points,

lines, circles, surfaces, and solids, as well as operations for translation, rotation, extrusion, intersection, subtraction, and composition.

A generated program could therefore describe a construction procedure rather than directly predicting the final geometry. The symbolic engine would execute this procedure and produce the corresponding geometric object.

This approach could potentially provide an explicit and editable representation of generated designs. In particular, the generated program could be inspected, modified, or reused by a human designer. The same principle could potentially apply to both two-dimensional and three-dimensional CAD systems.

8.3. Robotics

Robotics provides a further opportunity to investigate the combination of neural generation and symbolic execution. A robotics DSL could represent perception, spatial reasoning, planning, and actions through typed primitives.

For example, a program could combine operations for identifying an object, computing a position, selecting a trajectory, and executing an action. The symbolic component could enforce constraints on the available actions and the state of the environment, while the neural component could propose candidate action programs.

Such a representation could provide a natural interface between learned decision-making and explicit task-level plans. The resulting programs could also serve as interpretable descriptions of the actions selected by the system.

8.4. Autonomous Driving

A similar architecture could be considered for autonomous driving, provided that the DSL is carefully designed around the safety and temporal constraints of the domain.

A domain-specific language could represent operations such as identifying objects, maintaining a representation of the surrounding environment, evaluating possible trajectories, and selecting actions subject to explicit constraints.

Rather than directly producing low-level control signals, a neural model could generate structured decision programs that are subsequently validated and executed by a symbolic component. This separation could make it possible to inspect the intermediate decision process and to explicitly enforce domain-specific constraints.

However, autonomous driving is substantially more safety-critical than the domains considered in the present work. Such an application would therefore require substantially stronger validation, formal guarantees, simulation, and redundancy before deployment in a real system.

8.5. Natural Language and Programmatic Reasoning

Natural language is another possible application domain. Instead of defining a DSL directly over linguistic surface forms, the approach could employ a domain-specific intermediate language whose primitives represent semantic operations, transformations, or interactions with external tools.

The neural model could generate structured programs expressed in this intermediate representation, while a symbolic interpreter would execute the corresponding operations.

This perspective is particularly interesting because it provides a possible bridge between natural-language interfaces and executable symbolic representations. A natural-language instruction could be transformed into a program whose individual operations remain explicit and independently verifiable.

8.6. A General Domain Adaptation Strategy

These examples suggest a general methodology for transferring the proposed architecture to new domains. Let $\mathcal{D}$ denote a target domain and let $\mathcal{L}_{\mathcal{D}}$ denote a DSL designed for that domain. The adaptation process can be summarized as follows:

1. define a vocabulary of domain-specific primitives;
2. associate each primitive with explicit input and output types;
3. provide executable semantics for each primitive;
4. define a representation of programs suitable for neural processing;
5. define an evaluation function for generated programs; and
6. train the neural component to generate programs in $\mathcal{L}_{\mathcal{D}}$.

The neural architecture can consequently remain largely independent of the application domain, while the DSL and its execution engine provide the domain-specific knowledge.

This separation is a central research question for future work. An important objective would be to determine to what extent improvements to the general program-generation mechanism transfer across domains, and to what extent each new domain requires dedicated architectural or training modifications.

8.7. Towards Composable Domain-Specific Intelligence

A longer-term direction is to investigate whether several DSLs could be combined into a hierarchy of increasingly general abstractions. A system could, for example, combine geometric reasoning, physical interaction, and high-level planning primitives in a robotics setting.

Such a hierarchy would make it possible to investigate whether a neural model can learn to compose heterogeneous symbolic operations while relying on the execution engine to maintain domain-specific constraints.

More generally, this suggests viewing the DSL not merely as a restricted programming language, but as an explicit interface between learned models and executable domain knowledge.

9. Conclusion

We have presented a neuro-symbolic architecture for program synthesis that learns a semantic latent space of DSL programs purely from a cost-guided objective. The resulting embeddings self-organise into interpretable clusters without any explicit semantic labels, and the inference procedure exhibits emergent meta-cognitive search strategies.

These results suggest that typed DSLs combined with multi-component cost functions and iterative online training offer a promising path toward **explainable, data-efficient program synthesis** – addressing two of the most under-explored areas in neuro-symbolic AI: explainability and meta-cognition.

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Appendix A. Hyperparameters

Table 2: Model hyperparameters.

Component	Parameter	Value
GNN	layers	3
GNN	hidden dim	128
CNN	layers	4
CNN	channels	64→64→128→128
Transformer	decoder layers	4
Transformer	d_{model}	256
Transformer	heads	8
Training	learning rate	10^{-4}
Inference	beam width	20
Inference	max program length	128 tokens

Appendix B. Example Search Trace

B.1. Extract of dsl_model.md

The following trace illustrates the iterative cost-guided search.

1/961 Target program: `trim(I)` (trajectory: 93 programs)

- 2026-08-26 09:58:32.355692 #0(0)/1 Searched program (1/93): `switch(I, ONE, TWO)`, cost: 3338.632368475567

– Program pool

- * Program: `I`, cost: 3350.632368475567
- * Program: `I`, cost: 3350.632368475567
- * Program: `I`, cost: 3350.632368475567
- * Program: `I`, cost: 3350.632368475567
- * Program: `I`, cost: 3350.632368475567
- * Program: `I`, cost: 3350.632368475567
- * Program: `I`, cost: 3350.632368475567
- * Program: `I`, cost: 3350.632368475567
- * Program: `I`, cost: 3350.632368475567
- * Program: `I`, cost: 3350.632368475567

- Found worst program: `vmirror(I)`, cost: 3350.632368475567
- Found worst program: `lefthalf(I)`, cost: 3393.2498693133093
- Found worst program: `rot270(I)`, cost: 3377.604177212805
- Found worst program: `hmirror(I)`, cost: 3350.632368475567
- Found worst program: `dedupe(I)`, cost: 3350.632368475567

- **Found worst program:** cmirror(I), cost: 3377.604177212805
 - **Found worst program:** compress(I), cost: 3350.632368475567
 - **Found worst program:** branch(T, I, I), cost: 3350.632368475567
 - **Found worst program:** dmirror(I), cost: 3377.604177212805
 - **Found worst program:** righthalf(I), cost: 3430.2498693133093
 - **Found worst program:** vconcat(I, I), cost: 10004.2107936166
 - *Found better program:* trim(I), cost: 0.0 Found program: trim(I) (16(12)/95 iterations)
 - 2/961 Target program: rot90(I) (trajectory: 108 programs)
- 2026-08-26 09:58:33.619647 #0(0)/1 Searched program (1/108): vconcat(cmirror(I), canvas(SEVEN, increment(UP))), cost: 2568.1364918359345
 - Program pool
 - * Program: I, cost: 2576.6696437031455
 - * Program: I, cost: 2576.6696437031455
 - * Program: I, cost: 2576.6696437031455
 - * Program: I, cost: 2576.6696437031455
 - * Program: I, cost: 2576.6696437031455
 - * Program: I, cost: 2576.6696437031455
 - * Program: I, cost: 2576.6696437031455
 - * Program: I, cost: 2576.6696437031455
 - * Program: I, cost: 2576.6696437031455
 - * Program: I, cost: 2576.6696437031455
 - * Program: I, cost: 2576.6696437031455
 - **Found worst program:** crop(hmirror(hmirror(I)), UNITY, UP), cost: 6355.92407534942
 - **Found worst program:** hmirror(hmirror(dedupe(I))), cost: 5108.635176843368
 - **Found worst program:** crop(hmirror(dedupe(I)), ORIGIN, UP), cost: 6355.92407534942
 - **Found worst program:** crop(hmirror(hmirror(I)), toivec(TWO), UNITY), cost: 6339.986342171008
 - **Found worst program:** hmirror(switch(lefthalf(I), decrement(ONE), ONE)), cost: 4290.497533825093
 - **Found worst program:** switch(hmirror(hmirror(I)), TWO, ONE), cost: 2680.6696437031455
 - *Found better program:* rot270(hmirror(vmirror(I))), cost: 0.0 Found program: rot270(hmirror(vmirror(I))) (8(7)/52 iterations)
 - 3/961 Target program: dedupe(I) (trajectory: 563 programs)
 - 2026-08-26 09:58:34.388240 #0(0)/1 Searched program (1/563): cover(rot270(I), dneighbors(lrcorner(neighbors(UP)))), cost: 2672.8400280289097
 - Program pool
 - * Program: I, cost: 2697.4256661510863
 - * Program: I, cost: 2697.4256661510863
 - * Program: I, cost: 2697.4256661510863

- * Program: I, cost: 2697.4256661510863
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- * Program: I, cost: 2697.4256661510863
- * Program: I, cost: 2697.4256661510863
- * Program: I, cost: 2697.4256661510863
- *Found better program:* crop(hmirror(hmirror(hmirror(I))), UNITY, UNITY), cost: 1411.3630833845389
- 2026-08-26 09:58:34.438827 #1(1)/2 Searched program (373/563): crop(I, UNITY, UP), cost: 1411.293765408777
- Program pool
 - * Program: I, cost: 2697.4256661510863
 - * Program: I, cost: 2697.4256661510863
 - * Program: I, cost: 2697.4256661510863
 - * Program: I, cost: 2697.4256661510863
 - * Program: I, cost: 2697.4256661510863
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 - * Program: I, cost: 2697.4256661510863
 - * Program: I, cost: 2697.4256661510863
 - * Program: crop(hmirror(hmirror(hmirror(I))), UNITY, UNITY), cost: 1411.3630833845389
- **Found worst program:** dmirror(I), cost: 2707.8400280289097
- **Found worst program:** cmirror(I), cost: 2707.8400280289097
- *Found better program:* dedupe(I), cost: 0.0

B.2. GPT-5

Input prompt: <https://github.com/Julien-Livet/aicpp/blob/dev/scripts/data/gpt/gpt-5/training/1caeab9d-input-002.md>

Output: <https://github.com/Julien-Livet/aicpp/blob/dev/scripts/data/gpt/gpt-5/training/1caeab9d-output-002.md>

B.3. gpt-oss:120b

Input prompt: https://github.com/Julien-Livet/aicpp/blob/dev/scripts/data/ollama/gpt-oss_120b-cloud/training/1caeab9d-input-005.md

Output: https://github.com/Julien-Livet/aicpp/blob/dev/scripts/data/ollama/gpt-oss_120b-cloud/training/1caeab9d-output-005.md